Track A: model and verification

Scope: all 3 supplied S demand vectors, capacity-constrained clustering only. Seed-generated inputs (A2) remain a submission gap. Quantum optimization returns groups; all routing is exact classical evaluation with those groups fixed. No quantum speedup claim.

Binary x_ik assigns each customer exactly once. Minimize the sum of Euclidean customer-pair distances within each vehicle, subject to load <= capacity. Unused vehicles are allowed. The depot is fixed at (50, 50) and is absent from compactness.

The binary-slack target QUBO divides distances by the largest customer distance, adds assignment squares and binary-slack capacity squares. With U the normalized certificate cost, A = B = U + 1 proves feasibility of every target ground state. The certificate is not a solver warm start. Pair coefficients are twice off-diagonal matrix entries.

Variable counts (binary slack / unbalanced): S 14 / 8; M 45 / 30. Unbalanced H adds sum_k(-lambda1*h_k + lambda2*h_k^2), h_k = Q - sum_j d_j*x_jk, to normalized compactness and assignment squares. No slack bits. Its weights can change the optimum; the slack feasibility proof does not apply.

Evidence: 3 exact clustering/full-CVRP reference pairs; 3 QUBO validations (3 passed); methods: {'exhaustive-bitstrings': 3}. S checks all 14-bit strings; M uses the penalty proof and every feasible partition, not all 2^45 strings. Reported independent CPLEX checks agree on 3/3 instances.

The separate docs/MODEL.md gives the original CVRP arc/assignment formulation, connectivity constraints, penalty expansion and native Hamiltonian derivation.

Instances, embedding and pulse

3 explicit S demand vectors with (N,K,Q)=[(4, 2, 7)]. The selected input file contains N, K, Q, demand_range, demands, customer coordinates and depot. Fixed coordinates are read directly from the input. Feasibility witnesses are computed internally, not supplied as input.

References use exhaustive exact search: the Python API uses subset dynamic programming and Held-Karp tours; the S notebook also enumerates assignments and tour orders. M uses subset DP to avoid factorial tour enumeration. Capacity-filtered partitions select groups and tours jointly for full CVRP, independently of compactness. Vehicle labels are quotiented. S additionally counts tour optima modulo reversal.

Embedding uses a legal triangular grid, seeded greedy atom-label swaps and a least-squares positive interaction scale. Every pair is retained, including unwanted interactions where the target is zero. Device validation enforces 5 um spacing, 38 um radius, 80 atoms and all installed AnalogDevice controls.

Native energy at zero drive is -delta*sum(z) + sum(C6/r^6*z_a*z_b). C6 = 865723.02 rad/us*um^6 from Pulser. The fitted scale and scaled constant are saved. Global detuning fits the mean diagonal only: this is not an exact QUBO.

Pair error is ||J_real - s J_target||_F / ||s J_target||_F over all off-diagonal pairs. Linear error is ||h_real - s h||_2 / ||s h||_2. Embedding fidelity is characterized by these errors, not by a claimed exact mapping.

Calibration: {'duration_ns': 1000, 'amplitude': 8.0, 'initial_detuning': -8.0}. Status: completed; fallback: None. Candidate sweep durations, amplitudes and initial detunings are recorded in calibration trials; the final amplitude is zero. Settings freeze after the first instance. Calibration uses a benchmark instance, so it is not held-out validation.

Recorded quantum backends: {'emu-sv': 6}. S uses emu-sv; M uses emu-mps on CPU. MPS limits (maximum bond dimension, precision): []. Time steps, cutoffs, seeds and resource limits are recorded per run. MPS is approximate; convergence is not established. L, DMM and QPU operation are deferred.

Observed results and comparison

Quantum run statuses: {'quantum-global: completed': 3, 'quantum-unbalanced: completed': 3}. All statuses, failure messages, actual shots, reference values and gaps are preserved in results.csv.

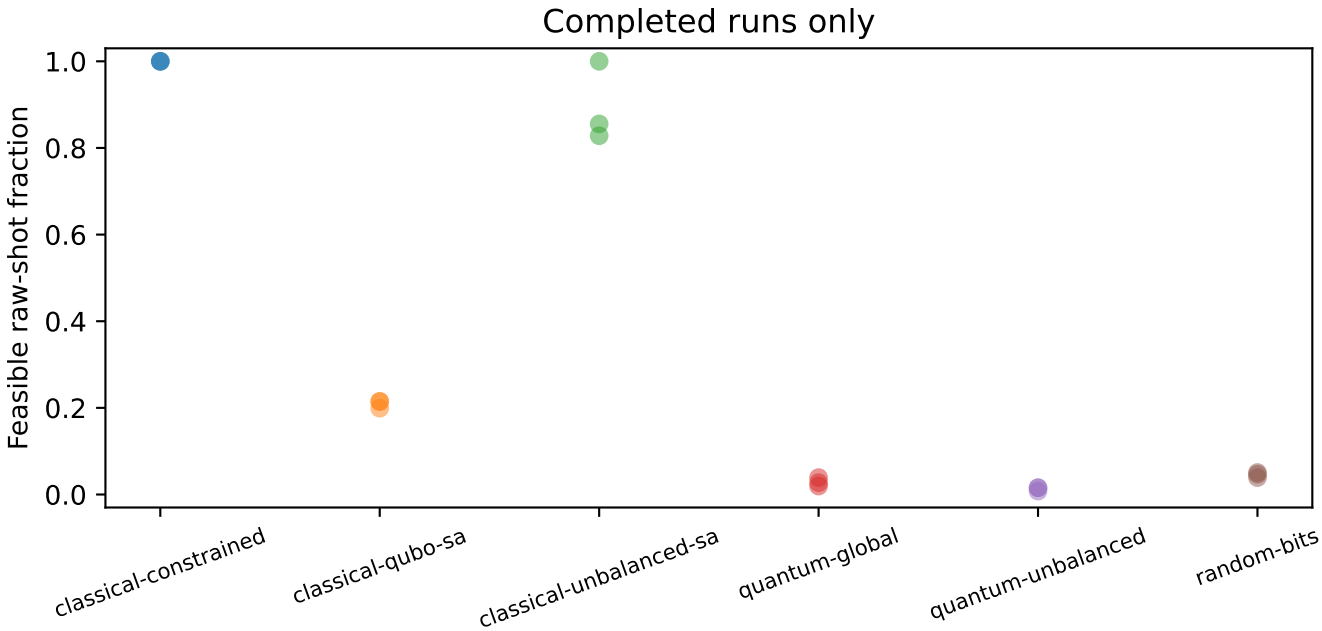
Additional unbalanced runs: 6; (lambda1, lambda2): [(0.2, 0.01)]; assignment penalty range: (2.1769122639957788, 3.301381342702836). Compare original feasible costs, not energies across formulations. Its separate pulse calibration is saved in unbalanced/calibration.json.

Feasibility is measured against the original assignment and capacity constraints, not against slack-bit residuals. Reference-hit probabilities use all raw shots as denominator. Gap thresholds: 0.1%, 1%, 5%. Undefined gaps remain missing; negative best-known gaps are never clipped.

The best compactness gap and best CVRP gap may come from different sampled partitions. Each receives exact fixed-group depot tours; routing never changes groups or feeds back into the quantum optimizer.

Classical constrained results are deterministic exact solutions repeated to the common shot budget, not independent stochastic samples. Classical QUBO comparisons use independent Metropolis chains with recorded schedules; random baselines use uniform raw bits. Both use identical decoding and feasibility filtering.

Reference, clustering, embedding, emulation and routing runtimes are separated. Reference DP is offline. The deterministic clustering baseline independently enumerates feasible partitions; this comparison measures quality rather than speedup.



What did not work, limits and AI use

The dense slack QUBO cannot generally be reproduced by 2D repulsive interactions with one shared detuning. Positive coupling signs alone do not establish embeddability. Large saved errors are physical-model approximation failures, not failures of the proven target penalty or exact references.

Completed quantum runs: 6; runs with no feasible raw shots: 0. L is intentionally out of scope. Resource-limited or failed runs have no reported success metrics.

Resource limits bound each isolated emulator by wall time and address space. MPS bond/truncation convergence, time-step convergence and noise robustness are not established by this baseline. No scaling or hardware performance claims are made.

Only one geometry is covered. Explicit demand vectors include different load ratios, but do not replace a seed-based instance generator. Low feasible rates on uniform bitstrings are expected for one-hot assignment. A small pulse calibration and local swap layout are baselines, not an exhaustive embedding/control optimization. There is no classical repair of quantum shots.

Reproduction: run the classical, quantum, then results notebooks with the .venv kernel, or call the shared Python experiment stages. Saved evidence is reassembled by the results notebook without rerunning clustering or quantum jobs. The three notebooks show executable core algorithms with beginner-friendly explanations and saved-result views.

AI-use disclosure: an AI assistant using Copilot SDK in VS Code implemented the shared modules, notebooks, derivations and automated report. The submitted numbers come from actual local solvers/emulators and saved raw outputs, not generated claims. Human review remains necessary.